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Physics 220 - Homework Assignment 6
Student Name: ________________ | Due Date: Fall 2024

Electromagnetic Waves and Optics Problems

Problem 1

A plane electromagnetic wave travels in free space in the +z direction. The electric field is given by:

$E(x,y,z,t) = E_0 \sin(kz - \omega t)\ \hat{i}$

where $E_0 = 100$ V/m, $k = 2\pi/\lambda$, and the frequency is 1.5×10^{15} Hz.

a) Find the wavelength λ and the angular frequency ω.

b) Write the expression for the magnetic field B.

c) Find the magnitude of the Poynting vector and determine the direction of energy flow.

Solution:

a) The frequency $f = 1.5 \times 10^{15}$ Hz

$\omega = 2\pi f = 2\pi \times 1.5 \times 10^{15} = 9.42 \times 10^{15}$ rad/s

For electromagnetic waves in vacuum: $c = \lambda f$, where $c = 3.00 \times 10^8$ m/s

$\lambda = c/f = (3.00 \times 10^8)/(1.5 \times 10^{15}) = 2.00 \times 10^{-7}$ m = 200 nm

$k = 2\pi/\lambda = 2\pi/(2.00 \times 10^{-7}) = 3.14 \times 10^7$ rad/m

b) For a plane wave in vacuum, E and B are perpendicular and related by:

$B = (1/c)(\hat{k} \times E)$

Since E is in the $\hat{i}$ direction and propagation is in $+\hat{k}$ direction:

$B = (E_0/c) \sin(kz - \omega t)\ \hat{j}$

$B_0 = E_0/c = 100/(3.00 \times 10^8) = 3.33 \times 10^{-7}$ T

Therefore: $B = (3.33 \times 10^{-7}\ \text{T}) \sin(kz - \omega t)\ \hat{j}$

c) The Poynting vector $S = (1/\mu_0)(E \times B)$

Since E is in $\hat{i}$ direction and B is in $\hat{j}$ direction, $E \times B$ is in $\hat{k}$ direction

$|S| = (1/\mu_0)|E \times B| = (1/\mu_0)E_0B_0 = (E_0^2)/(\mu_0 c)$

Using $\mu_0 = 4\pi \times 10^{-7}$ N/A²:

$|S| = (100)^2/((4\pi \times 10^{-7})(3.00 \times 10^8)) = 26.5$ W/m²

The Poynting vector points in the +z direction (direction of propagation).

Problem 2

Light of wavelength 600 nm is incident normally on a diffraction grating with 500 lines per mm.

a) Calculate the angle of the first-order maximum.

b) What is the maximum number of orders that can be observed?

c) If the same grating is used with light of wavelength 400 nm, how does the angular separation between orders change?

Solution:

a) The grating equation is: $d \sin \theta = m\lambda$

where d is the spacing between lines, m is the order, and θ is the diffraction angle.

$d = 1/(500 \text{ lines/mm}) = 1/(500 \times 10^3 \text{ lines/m}) = 2.00 \times 10^{-6}$ m

For first order (m = 1): $\sin \theta_1 = \lambda/d = (600 \times 10^{-9})/(2.00 \times 10^{-6}) = 0.300$

$\theta_1 = \sin^{-1}(0.300) = 17.5°$

b) The maximum order occurs when $\sin \theta = 1$:

$m_{max} = d/\lambda = (2.00 \times 10^{-6})/(600 \times 10^{-9}) = 3.33$

Since m must be an integer, the maximum observable order is m = 3.

c) For λ = 400 nm:

$\sin \theta_1 = (400 \times 10^{-9})/(2.00 \times 10^{-6}) = 0.200$

$\theta_1 = \sin^{-1}(0.200) = 11.5°$

The angular separation is smaller for shorter wavelengths. The angular separation between adjacent orders is proportional to λ, so it decreases when wavelength decreases.

Problem 3

A double-slit interference pattern is observed on a screen 2.0 m away from the slits. The slits are separated by 0.10 mm, and the wavelength of light is 550 nm.

a) Calculate the distance between adjacent bright fringes on the screen.

b) If the entire setup is immersed in water (n = 1.33), what happens to the fringe spacing?

c) Determine the intensity at a point on the screen that is 5.0 cm from the central maximum, relative to the central maximum intensity.

Solution:

a) For double-slit interference, the position of bright fringes is:

$y_m = m\lambda L/d$

where L is the distance to screen, d is slit separation, and m is the order.

The spacing between adjacent fringes is:

$\Delta y = y_{(m+1)} - y_m = \lambda L/d$

$\Delta y = (550 \times 10^{-9})(2.0)/(0.10 \times 10^{-3}) = 0.011 \text{ m} = 11 \text{ mm}$

b) In water, the wavelength changes: $\lambda\_water = \lambda\_air/n = 550/1.33 = 414$ nm

$\Delta y\_water = \lambda\_water\ L/d = (414 \times 10^{-9})(2.0)/(0.10 \times 10^{-3}) = 8.3$ mm

The fringe spacing decreases because the wavelength is shorter in water.

c) The intensity in double-slit interference is:

$I = I_0 \cos^2(\pi dy/(\lambda L))$

where I_0 is the intensity at the central maximum.

At $y = 5.0$ cm $= 0.050$ m:

$I/I_0 = \cos^2(\pi(0.10 \times 10^{-3})(0.050)/((550 \times 10^{-9})(2.0)))$

$I/I_0 = \cos^2(14.3) = 0.94$

The intensity is 94% of the central maximum intensity.